

# Wireless Security and Configuration — Guided Notes

## ANSWER KEY

Unit 2.2.2 | CompTIA Network+ N10-009 Obj. 2.2 | N10-008 2.2

For instructor use / distribute after students complete the notes.

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### Question 1

WEP is broken because its \_\_\_\_\_ -bit initialization vector space is too small, causing IVs to \_\_\_\_\_. This allows an attacker to crack the key in under \_\_\_\_\_. WEP should \_\_\_\_\_ be deployed.

*Answer: 24-bit; repeat after enough traffic is captured; a minute (with freely available tools); never.*

### Question 2

WPA used \_\_\_\_\_ as its encryption protocol, which wrapped around the same \_\_\_\_\_ cipher as WEP. WPA2 replaced this with \_\_\_\_\_, which uses the \_\_\_\_\_ block cipher with 128-bit keys.

*Answer: TKIP (Temporal Key Integrity Protocol); RC4; AES-CCMP; AES (Advanced Encryption Standard).*

### Question 3

WPA3 replaces the PSK handshake with \_\_\_\_\_ (SAE), which makes offline brute-force attacks \_\_\_\_\_ because the attacker must \_\_\_\_\_ for each guess. WPA3 also provides \_\_\_\_\_, meaning captured traffic cannot be decrypted even if the password is later compromised.

*Answer: Simultaneous Authentication of Equals; impossible/impractical; interact with the AP; forward secrecy.*

### Question 4

WPA2/WPA3 Personal uses a \_\_\_\_\_ shared by all users. Enterprise uses \_\_\_\_\_ where each user authenticates with \_\_\_\_\_ credentials. The AP passes authentication to a \_\_\_\_\_ server which checks against a directory.

*Answer: Single pre-shared password; 802.1X; unique individual; RADIUS.*

### Question 5

Enterprise mode advantages over Personal include: (1) \_\_\_\_\_, (2) \_\_\_\_\_, (3) \_\_\_\_\_, and (4) \_\_\_\_\_.

*Answer: Individual user accountability; dynamic VLAN assignment per user; per-user encryption keys; centralized revocation (disable account = immediate removal from network).*

### Question 6

On 2.4 GHz, the only three non-overlapping channels are \_\_\_\_\_, \_\_\_\_\_, and \_\_\_\_\_. Using channel 3 causes \_\_\_\_\_ interference which is \_\_\_\_\_ than using the same channel as a neighbor because devices can't decode the signal cleanly but still wait for it.

*Answer: 1; 6; 11; adjacent channel (ACI); worse.*

### Question 7

A rogue AP is \_\_\_\_\_ connected to your network without authorization. An evil twin is \_\_\_\_\_ that mimics a legitimate network. Evil twin attacks are more dangerous against \_\_\_\_\_ mode networks because clients have no way to verify the AP's \_\_\_\_\_.

*Answer: Any access point; a deliberate attacker-controlled AP; Personal (PSK); identity/certificate.*

### Question 8 — Real World Application

REAL WORLD: A teacher reports that their laptop keeps connecting to a network called 'STAFF' but then has no internet access and gets strange certificate warnings. Other staff are unaffected. What attack is likely occurring and what steps do you take?

*Answer: This is likely an evil twin attack — an attacker has set up a rogue AP with the SSID 'STAFF' and a stronger signal than the legitimate AP. The certificate warning indicates the rogue AP's TLS certificate doesn't match the legitimate RADIUS server's certificate. Steps: (1) Check wireless controller for unknown BSSIDs broadcasting 'STAFF'. (2) Use a wireless analyzer to locate the rogue AP by signal strength. (3) Remove or disable the rogue device. (4) Verify the teacher's device is configured to validate the RADIUS server certificate and reject unknown certificates. (5) Ensure WPA3 transition mode is enabled — SAE makes this attack harder.*